Multiple Linear Model Code

```r
# Loads in library
library(car)
```

```
## Loading required package: carData
```

```r
# Sets seed for reproducibility
set.seed(42)
```

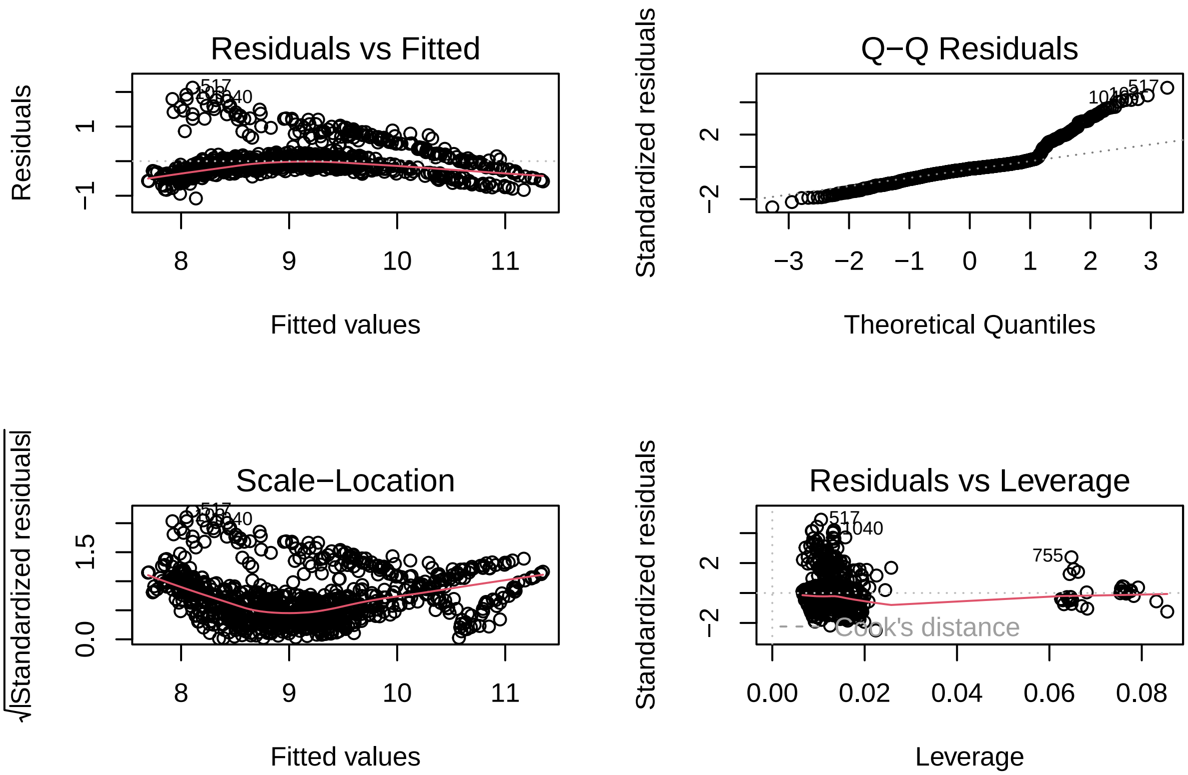
```r
# Reads in data frame
df <- read.csv("insurance.csv")

# Creates log transformation of charges
df$log_charges <- log(df$charges)

# Sets categorical variables to be factors
df$sex <- as.factor(df$sex)
df$children <- as.factor(df$children)
df$smoker <- as.factor(df$smoker)
df$region <- as.factor(df$region)

# Creates training and testing sets
n <- nrow(df)
train_index <- sample(1:n, size = 0.7 * n)

df_train <- df[train_index, ]
df_test  <- df[-train_index, ]

# Shows header of data frame
head(df)
```

```
##   age    sex    bmi children smoker    region   charges log_charges
## 1  19 female 27.900        0    yes southwest 16884.924    9.734176
## 2  18   male 33.770        1     no southeast  1725.552    7.453302
## 3  28   male 33.000        3     no southeast  4449.462    8.400538
## 4  33   male 22.705        0     no northwest 21984.471    9.998092
## 5  32   male 28.880        0     no northwest  3866.855    8.260197
## 6  31 female 25.740        0     no southeast  3756.622    8.231275
```

```r
# Creates regression model of all variables
lm1 <- lm(log_charges ~ smoker + age + children + bmi + region + sex,
          data=df_train)
# Prints model summary
summary(lm1)
```

```
##
## Call:
## lm(formula = log_charges ~ smoker + age + children + bmi + region +
##     sex, data = df_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.07789 -0.22189 -0.04631  0.08029  2.12366
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.991098   0.087136  80.232  < 2e-16 ***
## smokeryes        1.553319   0.035863  43.312  < 2e-16 ***
## age              0.034481   0.001030  33.469  < 2e-16 ***
## children1        0.158692   0.036189   4.385 1.29e-05 ***
## children2        0.311497   0.040708   7.652 4.97e-14 ***
## children3        0.262875   0.046483   5.655 2.07e-08 ***
## children4        0.508188   0.108366   4.690 3.15e-06 ***
## children5        0.384698   0.119182   3.228  0.00129 **
## bmi              0.014495   0.002462   5.887 5.49e-09 ***
## regionnorthwest -0.075781   0.041502  -1.826  0.06818 .
## regionsoutheast -0.172300   0.041070  -4.195 2.99e-05 ***
## regionsouthwest -0.166744   0.041214  -4.046 5.65e-05 ***
## sexmale         -0.072515   0.028760  -2.521  0.01186 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.436 on 923 degrees of freedom
## Multiple R-squared:  0.772,  Adjusted R-squared:  0.7691
## F-statistic: 260.5 on 12 and 923 DF,  p-value: < 2.2e-16
```

```r
# Prints model diagnostic plots
par(mfrow=c(2,2))
plot(lm1)
```




```r
# Calculates rmse for training set
sqrt(mean((df[as.numeric(names(lm1$fitted.values)),]$charges - exp(lm1$fitted.values))^2))
```

```
## [1] 8252.762
```

```r
# Calculates rmse for test set
pred <- exp(predict(lm1, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

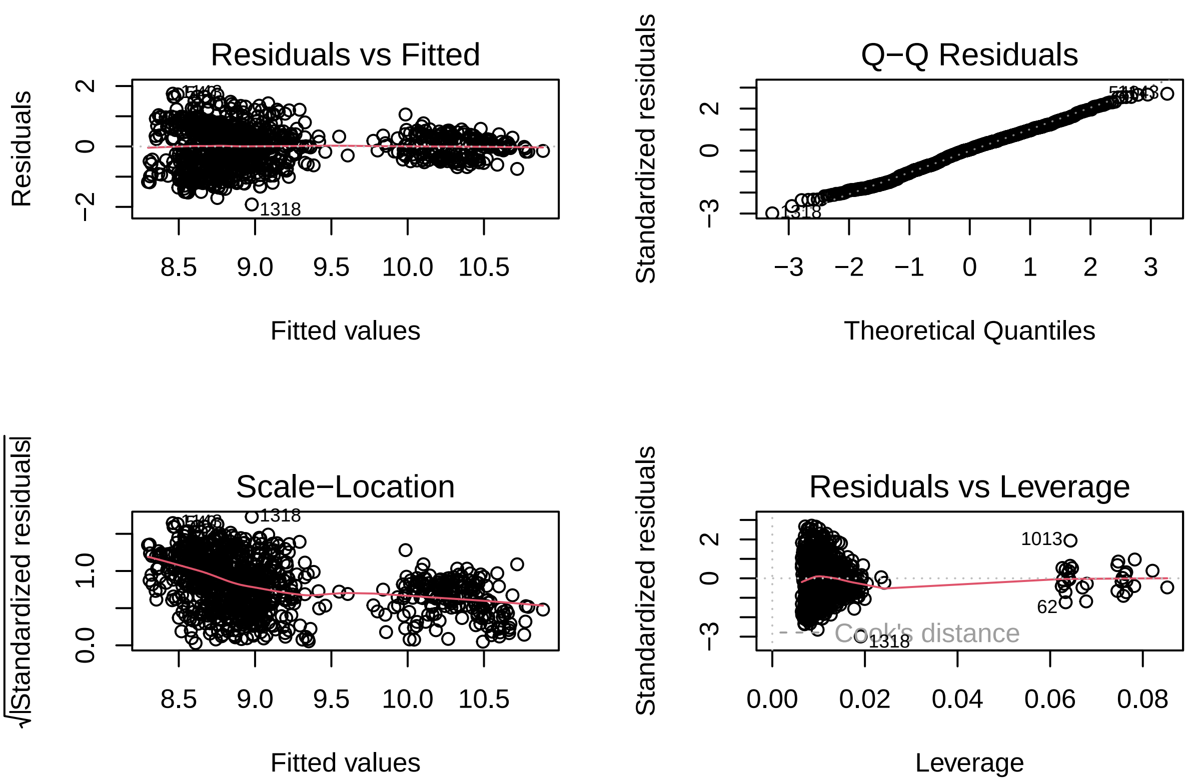
```
## [1] 8588.915
```

```r
# Creates regression model without age variable
lm2 <- lm(log_charges ~ smoker + bmi + children + region + sex, data=df_train)
# Prints model summary
summary(lm2)
```

```
##
## Call:
## lm(formula = log_charges ~ smoker + bmi + children + region +
##     sex, data = df_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.91934 -0.45623  0.03739  0.43883  1.74712
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8.136405   0.119163  68.280  < 2e-16 ***
## smokeryes        1.515291   0.053302  28.428  < 2e-16 ***
## bmi              0.022294   0.003645   6.117 1.41e-09 ***
## children1        0.185969   0.053800   3.457 0.000572 ***
## children2        0.340696   0.060520   5.630 2.40e-08 ***
## children3        0.380722   0.068923   5.524 4.31e-08 ***
## children4        0.642097   0.161032   3.987 7.21e-05 ***
## children5        0.250841   0.177126   1.416 0.157062
## regionnorthwest -0.090469   0.061710  -1.466 0.142982
## regionsoutheast -0.229807   0.061019  -3.766 0.000176 ***
## regionsouthwest -0.166679   0.061285  -2.720 0.006656 **
## sexmale         -0.112600   0.042729  -2.635 0.008549 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6483 on 924 degrees of freedom
## Multiple R-squared:  0.4954, Adjusted R-squared:  0.4894
## F-statistic: 82.46 on 11 and 924 DF,  p-value: < 2.2e-16
```

```r
# Prints diagnostic plots of model
par(mfrow=c(2,2))
plot(lm2)
```



```r
# Calculates rmse for training set
sqrt(mean((df[as.numeric(names(lm2$fitted.values)),]$charges - exp(lm2$fitted.values))^2))
```

```
## [1] 7275.045
```

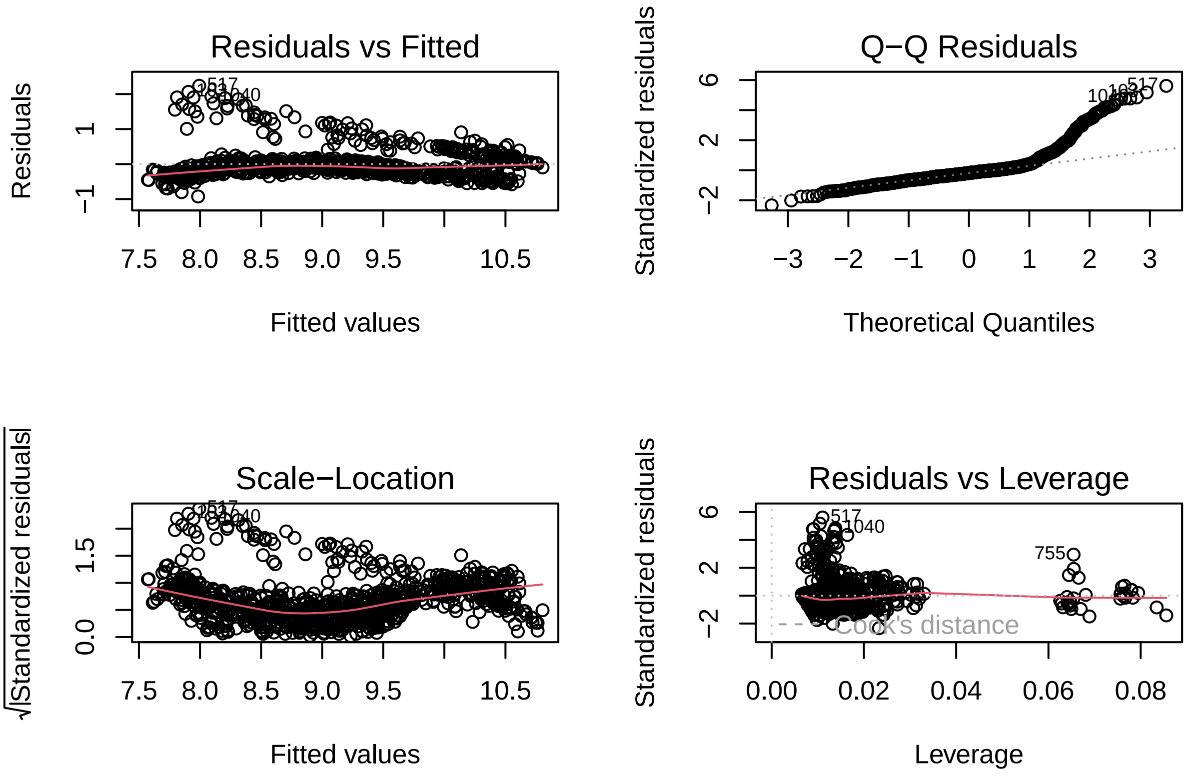
```r
# Calculates rmse for testing set
pred <- exp(predict(lm2, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

```
## [1] 7011.378
```

```r
# Creates regression with model with smoker and age interaction
lm3 <- lm(log(charges) ~ smoker * age + bmi + children + region + sex,
          data=df_train)
# Prints model summary
summary(lm3)
```

```
##
## Call:
## lm(formula = log(charges) ~ smoker * age + bmi + children + region +
##     sex, data = df_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92545 -0.20737 -0.07373  0.05195  2.23667
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.773572   0.081821  82.785  < 2e-16 ***
## smokeryes        2.753719   0.097749  28.171  < 2e-16 ***
## age              0.040586   0.001056  38.418  < 2e-16 ***
## bmi              0.013316   0.002265   5.879 5.77e-09 ***
## children1        0.164910   0.033272   4.956 8.54e-07 ***
## children2        0.320509   0.037429   8.563  < 2e-16 ***
## children3        0.262099   0.042732   6.134 1.27e-09 ***
## children4        0.477219   0.099649   4.789 1.95e-06 ***
## children5        0.418735   0.109595   3.821 0.000142 ***
## regionnorthwest -0.067665   0.038157  -1.773 0.076507 .
## regionsoutheast -0.160261   0.037767  -4.243 2.42e-05 ***
## regionsouthwest -0.168021   0.037888  -4.435 1.03e-05 ***
## sexmale         -0.066746   0.026442  -2.524 0.011764 *
## smokeryes:age   -0.030976   0.002375 -13.045  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4008 on 922 degrees of freedom
## Multiple R-squared:  0.8076, Adjusted R-squared:  0.8048
## F-statistic: 297.6 on 13 and 922 DF,  p-value: < 2.2e-16
```

```r
# Prints model diagnostic plots
par(mfrow=c(2,2))
plot(lm3)
```




```r
# Calculates rmse for training set
sqrt(mean((df[as.numeric(names(lm3$fitted.values)),]$charges - exp(lm3$fitted.values))^2))
```

```
## [1] 6262.846
```

```r
# Calculates rmse for testing set
pred <- exp(predict(lm3, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

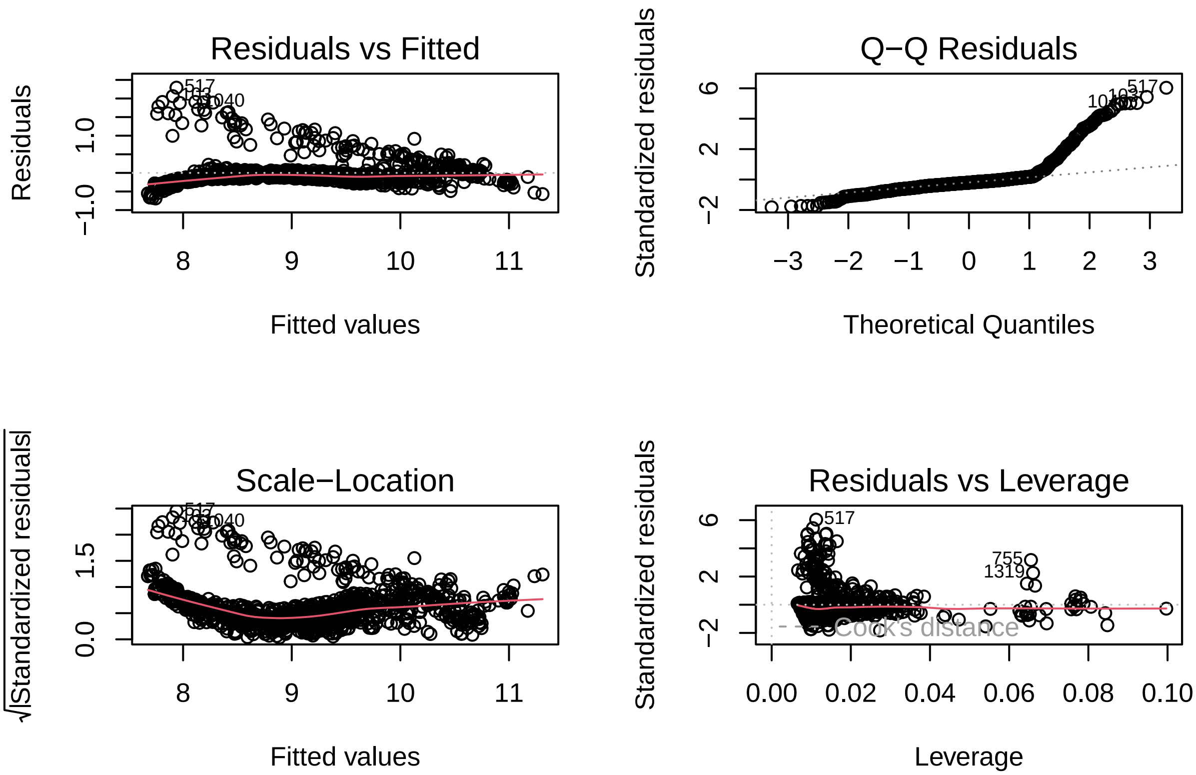
```
## [1] 5987.645
```

```r
# Creates regression model of smoker interaction with bmi and age
lm4 <- lm(log_charges ~ smoker * age + smoker * bmi + children + region + sex,
          data=df_train)
# Prints model summary
summary(lm4)
```

```
##
```

```
## Call:
## lm(formula = log_charges ~ smoker * age + smoker * bmi + children +
##     region + sex, data = df_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.68702 -0.15631 -0.07742  0.01336  2.29098
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.083937   0.084216  84.116  < 2e-16 ***
## smokeryes        1.254863   0.179906   6.975 5.83e-12 ***
## age              0.041079   0.001008  40.764  < 2e-16 ***
## bmi              0.002711   0.002417   1.121  0.26237
## children1        0.159058   0.031704   5.017 6.30e-07 ***
## children2        0.306750   0.035687   8.596  < 2e-16 ***
## children3        0.263381   0.040711   6.470 1.59e-10 ***
## children4        0.483771   0.094938   5.096 4.22e-07 ***
## children5        0.466752   0.104528   4.465 8.99e-06 ***
## regionnorthwest -0.065365   0.036354  -1.798  0.07250 .
## regionsoutheast -0.154216   0.035986  -4.285 2.02e-05 ***
## regionsouthwest -0.175240   0.036103  -4.854 1.42e-06 ***
## sexmale         -0.067005   0.025192  -2.660  0.00796 **
## smokeryes:age   -0.031685   0.002263 -13.998  < 2e-16 ***
## smokeryes:bmi    0.049621   0.005096   9.737  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3818 on 921 degrees of freedom
## Multiple R-squared:  0.8255, Adjusted R-squared:  0.8229
## F-statistic: 311.3 on 14 and 921 DF,  p-value: < 2.2e-16
```

```r
# Prints model diagnostic plots
par(mfrow=c(2,2))
plot(lm4)
```




```r
# Calculates rmse for training data
sqrt(mean((df[as.numeric(names(lm4$fitted.values)),]$charges - exp(lm4$fitted.values))^2))
```

```
## [1] 5645.03
```

```r
# Calculates rmse for testing data
pred <- exp(predict(lm4, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

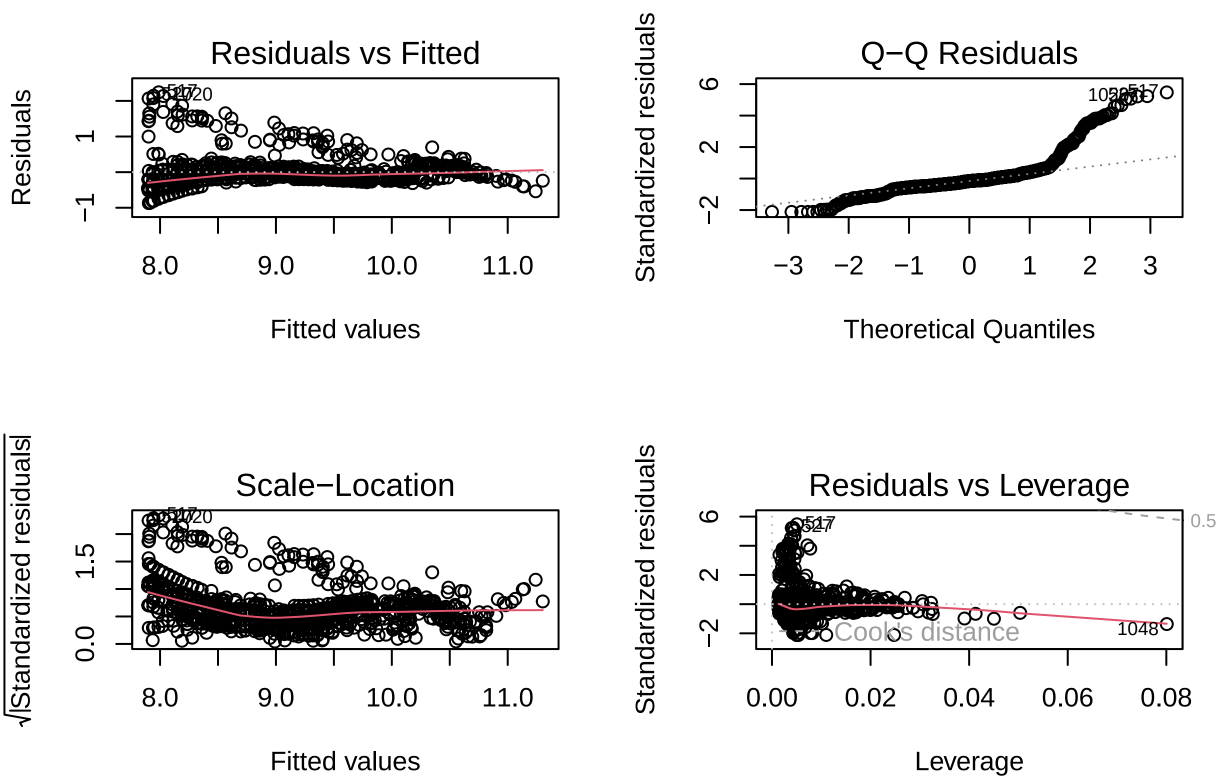
```
## [1] 5574.061
```

```r
# Creates model with just interaction terms
lm5 <- lm(log(charges) ~ smoker * bmi + smoker * age, data=df_train)
# Prints model summary
summary(lm5)
```

```
##
## Call:
## lm(formula = log(charges) ~ smoker * bmi + smoker * age, data = df_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.86830 -0.19169 -0.06477  0.06149  2.24178
```

```
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     7.1302313  0.0861507  82.765  < 2e-16 ***
## smokeryes       1.2262912  0.1931473   6.349 3.38e-10 ***
## bmi             0.0006657  0.0025151   0.265    0.791
## age             0.0417274  0.0010762  38.772  < 2e-16 ***
## smokeryes:bmi   0.0499702  0.0054650   9.144  < 2e-16 ***
## smokeryes:age -0.0316279  0.0024309 -13.011  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4107 on 930 degrees of freedom
## Multiple R-squared:  0.7962, Adjusted R-squared:  0.7951
## F-statistic: 726.5 on 5 and 930 DF,  p-value: < 2.2e-16
```

```r
# Prints model diagnostic plots
par(mfrow=c(2,2))
plot(lm5)
```




```r
# Calculates training rmse
sqrt(mean((df[as.numeric(names(lm5$fitted.values)),]$charges - exp(lm5$fitted.values))^2))
```

```
## [1] 5387.139
```

```r
# Calculates testing rmse
pred <- exp(predict(lm5, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

```
## [1] 4978.821
```